

CHAPTER 3

TRENDS IN DATA WAREHOUSING

CHAPTER OBJECTIVES

- Review the continued growth in data warehousing
- Learn how data warehousing has become mainstream
- Discuss several major trends, one by one
- Grasp the need for standards and review the progress
- Understand Web-enabled data warehouse

In the previous chapters, we have seen why business intelligence along with the underlying data warehouses is essential for enterprises of all sizes in all industries. We have reviewed how businesses are reaping major benefits from data warehousing. We have also discussed the building blocks of a data warehouse. You now have a fairly good idea of the features and functions of the basic components and a reasonable definition of data warehousing. You have understood that it is a fundamentally simple concept; at the same time, you know it is also a blend of many technologies. Several business and technological drivers have moved data warehousing forward in the past few years.

Before we proceed further, we are at the point where we want to ask some relevant questions. What is the current scenario and state of the market? What businesses have adopted data warehousing? What are the technological advances? In short, what are the significant trends?

Are you wondering if it is too early in our discussion of the subject to talk about trends? The usual practice is to include a chapter on future trends towards the end, almost as an afterthought. The reader typically glosses over the discussion on future trends. This chapter is not so much like looking into the crystal ball for possible future happenings; we want to deal with the important trends that are happening now.

It is important for you to keep the knowledge about current trends as a backdrop in your mind as you continue the in-depth study of the subject. When you gather the informational requirements for your data warehouse, you need to be aware of current trends. When you get into the design phase, you need to be cognizant of the trends. When you implement your data warehouse, you need to ensure that your data warehouse is in line with the trends. Knowledge of trends is important and necessary even at a fairly early stage of your study.

In this chapter, we will touch upon most of the major trends. You will understand how and why data warehousing continues to grow and become more and more pervasive. We will discuss trends in vendor solutions and products. We will relate data warehousing with other technological phenomena such as the Internet and the World Wide Web. Wherever more detailed discussions are necessary, we will revisit some of the trends in later chapters.

CONTINUED GROWTH IN DATA WAREHOUSING

Data warehousing is no longer a purely novel idea for study and experimentation. It has become mainstream. True, the data warehouse is not in every dentist's office yet, but neither is it confined only to high-end businesses. More than half of all U.S. companies have made a commitment to data warehousing. About 90% of multinational companies have data warehouses or are planning to implement data warehouses in the next few months.

In every industry across the board, from retail chain stores to financial institutions, from manufacturing enterprises to government departments, from airline companies to pharmaceuticals, data warehousing has revolutionized the way people perform business analysis and make strategic decisions. Every company that has a data warehouse continues to realize enormous benefits that get translated into positive results at the bottom line. Many of these companies, incorporating Web-based technologies, have enhanced the potential for greater and easier delivery of vital information.

Even during the first few years of data warehousing in the late 1990s, hundreds of vendors had flooded the market with numerous products. Vendor solutions and products run the gamut of data warehousing: data modeling, data acquisition, data quality, data analysis, metadata, and so on. A buyer's guide published by the Data Warehousing Institute at that time featured no fewer than 105 leading products. The market is huge and continues to grow in revenue dollars.

Data Warehousing has Become Mainstream

In the early stages, four significant factors drove many companies to move into data warehousing:

- Fierce competition
- Government deregulation
- Need to revamp internal processes
- Imperative for customized marketing

Telecommunications, banking, and retail were the first industries to adopt data warehousing. That was largely because of government deregulation in telecommunications and banking. Retail businesses moved into data warehousing because of fiercer competition. Utility companies joined the group as that sector was deregulated. The next wave of

businesses to get into data warehousing consisted of companies in financial services, health care, insurance, manufacturing, pharmaceuticals, transportation, and distribution.

Today, telecommunications and financial services continue to lead in data warehouse spending. As much as 15% of technology budgets in these industries is spent on data warehousing. Companies in these industries collect large volumes of transaction data. Data warehousing is able to transform such large volumes of data into strategic information useful for decision making.

At present, data warehouses exist in every conceivable industry. Figure 3-1 lists the industries in the order of the average salaries paid to data warehousing professionals. In 2008, pharmaceuticals was at the top of the list, with the highest average salary.

In the early stages of data warehousing, it was, for the most part, used exclusively by global corporations. It was expensive to build a data warehouse and the tools were not quite adequate. Only large companies had the resources to spend on the new paradigm. Now we have begun to see a strong presence of data warehousing in medium-sized and smaller companies, which are now able to afford the cost of building data warehouses or buying turnkey data marts. Take a look at the database management systems (DBMSs) you have been using in the past. You will find that the database vendors have now added features to assist you in building data warehouses using these DBMSs. Packaged solutions have also become less expensive and operating systems robust enough to support data warehousing functions.

Data Warehouse Expansion

Although earlier data warehouses concentrated on keeping summary data for high-level analysis, we now see larger and larger data warehouses being built by different businesses. Now companies have the ability to capture, cleanse, maintain, and use the vast amounts of data generated by their business transactions. The quantities of data kept in data warehouses continue to swell to the terabyte range. Data warehouses storing several terabytes of data are not uncommon in retail and telecommunications.

For example, take the telecommunications industry. A telecommunications company generates hundreds of millions of call-detail transactions in a year. For promoting the

Annual average salary in \$ 000			
Pharmaceuticals	117	Manufacturing (non-computer)	100
Consulting/professional	112	Telecommunications	98
Software/Internet	107	Healthcare	98
Government (federal)	105	Retail/wholesale/distribution	92
Financial Services	105	Insurance	90
Media/entertainment/publishing	103	Education	85
Computer manufacturing	103	Government (state/local)	82

Source: 2008 Data Warehousing Salary Survey by the Data Warehousing Institute

Figure 3-1 Industries using data warehousing.

proper products and services, the company needs to analyze these detailed transactions. The data warehouse for the company has to store data at the lowest level of detail.

Similarly, consider a retail chain with hundreds of stores. Every day, each store generates many thousands of point-of-sale transactions. Again, another example is a company in the pharmaceutical industry that processes thousands of tests and measurements for getting product approvals from the government. Data warehouses in these industries tend to be very large.

Finally, let us look at the potential size of a typical Medicaid fraud control unit of a large state. This organization is exclusively responsible for investigating and prosecuting health care fraud arising out of billions of dollars spent on Medicaid in that state. The unit also has to prosecute cases of patient abuse in nursing homes and monitor fraudulent billing practices by physicians, pharmacists, and other health care providers and vendors. Usually there are several regional offices. A fraud scheme detected in one region must be checked against all other regions. Can you imagine the size of the data warehouse needed to support such a fraud control unit? There could be many terabytes of data.

Vendor Solutions and Products

As an information technology professional, you are familiar with database vendors and database products. In the same way, you are familiar with most of the operating systems and their vendors. How many leading database vendors are there? How many leading vendors of operating systems are there? A handful? The number of database and operating system vendors pales in comparison with data warehousing products and vendors. There are hundreds of data warehousing vendors and thousands of data warehousing products and solutions.

In the beginning, the market was filled with confusion and vendor hype. Every vendor, small or big, that had any product remotely connected to data warehousing jumped on the bandwagon. Data warehousing meant what each vendor defined it to be. Each company positioned its own products as the proper set of data warehousing tools. Data warehousing was a new concept for many of the businesses that adopted it. These businesses were at the mercy of the marketing hype of the vendors.

Over the past decade, the situation has improved tremendously. The market is reaching maturity, to the extent of producing off-the-shelf packages and becoming increasingly stable. Figure 3-2 shows the current state of the data warehousing market as compared to how the market was in the beginning stages.

What do we normally see in any maturing market? We expect to find a process of consolidation. And that is exactly what is taking place in the data warehousing market. Data warehousing vendors are merging to form stronger and more viable companies. Some major players in the industry are extending the range of their solutions by acquisition of other companies. Some vendors are positioning suites of products, their own or ones from groups of other vendors, piecing them together as integrated data warehousing solutions.

Now the traditional database companies are also in the data warehousing market. They have begun to offer data warehousing solutions built around their database products. On the one hand, data extraction and transformation tools are packaged with the database management system. On the other hand, inquiry and reporting tools are enhanced for data warehousing. Some database vendors take the enhancement further by offering sophisticated products such as data mining tools.

With so many vendors and products, how can we classify the vendors and products, and thereby make sense of the market? It is best to separate the market broadly into two distinct

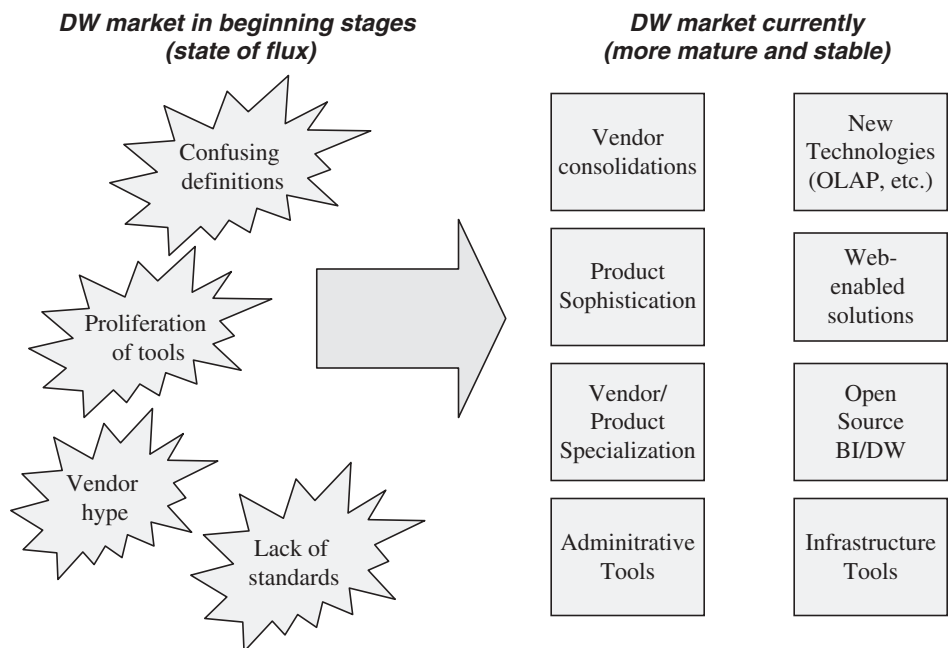


Figure 3-2 Status of the data warehousing market.

Administration & Operations: System/network administration, backup/restore, disaster recovery, performance/usage management, capacity planning, security, database management

Analytics: Business Activity Monitoring, Query/Reporting, OLAP, Forecasting, Data Mining, Dashboards/Scorecards, Visualization, Text Analysis, Analytic Application Development

BI Services: Analytic Service Providers (ASP), BI Software, Consultants, Systems Integrators, Industry Associations/Consortia, Research

Data Integration: Business Process Management, Data Profiling/Cleaning/Conversion/Movement, Data Mapping/Transformation, Master Data Management, Metadata Management

DW Design: Data Analysis, Data Modeling (Dimensional Modeling/Entity-Relationship Modeling), Data Warehousing Toolset

Information Delivery: Broadcasting, Collaboration, Content Management, Document Management, Enterprise Information Portals, Wireless Data Analysis

Infrastructure: Data Accelerators, Data Warehouse Appliances, Database Management Systems, Multidimensional Databases, Servers, Storage Management Systems

Open Source BI/DW: Business Intelligence, Data Integration, Databases, Consultants, Systems Integrators

Source: The Data Warehousing Institute

Figure 3-3 Categories of products and services.

groups. The first group consists of data warehouse vendors and products catering to the needs of corporate data warehouses in which all enterprise data is integrated and transformed. This segment has been referred to as the market for strategic data warehouses. This segment accounts for about a quarter of the total market. The second segment is more loose and dispersed, consisting of departmental data marts, fragmented database marketing systems, and a wide range of decision support systems. Specific vendors and products dominate each segment.

We may also look at the list of products in another way. Figure 3-3 shows a list of the types of data warehouse and business intelligence products and services, categorized based on the features and functions they perform.

SIGNIFICANT TRENDS

Some experts feel that, until now, technology has been driving data warehousing. These experts declare that we are now beginning to see important progress in software. In the next few years, data warehousing is expected make big strides in software, especially for optimizing queries, indexing very large tables, enhancing SQL, improving data compression, and expanding dimensional modeling.

Let us separate out the significant trends and discuss each briefly. Be prepared to visit each trend, one by one—each has a serious impact on data warehousing. As we walk through each trend, try to grasp its significance and be sure that you perceive its relevance to your company's data warehouse. Be prepared to answer the question: What must you do to take advantage of the trend in your data warehouse?

Real-Time Data Warehousing

Business intelligence systems and the supporting data warehouses have been used mainly for strategic decision making. The data warehouse was kept separate from operational systems. Recently industry momentum is swinging towards using business intelligence for tactical decision making for day-to-day business operations. Data warehousing is progressing rapidly to the point that real-time data warehousing is the focus of senior executives.

Traditional data warehousing is passive, providing historical trends, whereas real-time data warehousing is dynamic, providing the most up-to-date view of the business in real time. A real-time data warehouse gets refreshed continuously, with almost zero latency.

Real-time information delivery increases productivity tremendously by sharing information with more people. Companies are, therefore, coming under a lot of pressure to provide information, in real time, to everyone connected to critical business processes. However, extraction, transformation, and integration of data for real-time data warehousing have several challenges. We will discuss these in detail in a later chapter.

Multiple Data Types

When you build the first iteration of your data warehouse, you may include just numeric data. But soon you will realize that including structured numeric data alone is not enough. Be prepared to consider other data types as well.

Traditionally, companies included structured data, mostly numeric, in their data warehouses. From this point of view, decision support systems were divided into two

campus: data warehousing dealt with structured data, knowledge management involved unstructured data. This distinction is being blurred. For example, most marketing data consists of structured data in the form of numeric values. Marketing data also contains unstructured data in the form of images. Let us say a decision maker is performing an analysis to find the top-selling product types. The decision maker arrives at a specific product type in the course of the analysis. He or she would now like to see images of the products in that type to make further decisions. How can this be made possible? Companies are realizing there is a need to integrate both structured and unstructured data in their data warehouses.

What are the types of data we call unstructured data? Figure 3-4 shows the different types of data that need to be integrated in the data warehouse to support decision making more effectively.

Let us now turn to the progress made in the industry for including some of the types of unstructured data. You will gain an understanding of what must be done to include these data types in your data warehouse.

Adding Unstructured Data Some vendors are addressing the inclusion of unstructured data, especially text and images, by treating such multimedia data as just another data type. These are defined as part of the relational data and stored as binary large objects (BLOBs) up to 2 GB in size. User-defined functions (UDFs) are used to define these as user-defined types (UDTs).

Not all BLOBs can be stored simply as another relational data type. For example, a video clip would require a server supporting delivery of multiple streams of video at a given rate and synchronization with the audio portion. For this purpose, specialized servers are being provided.

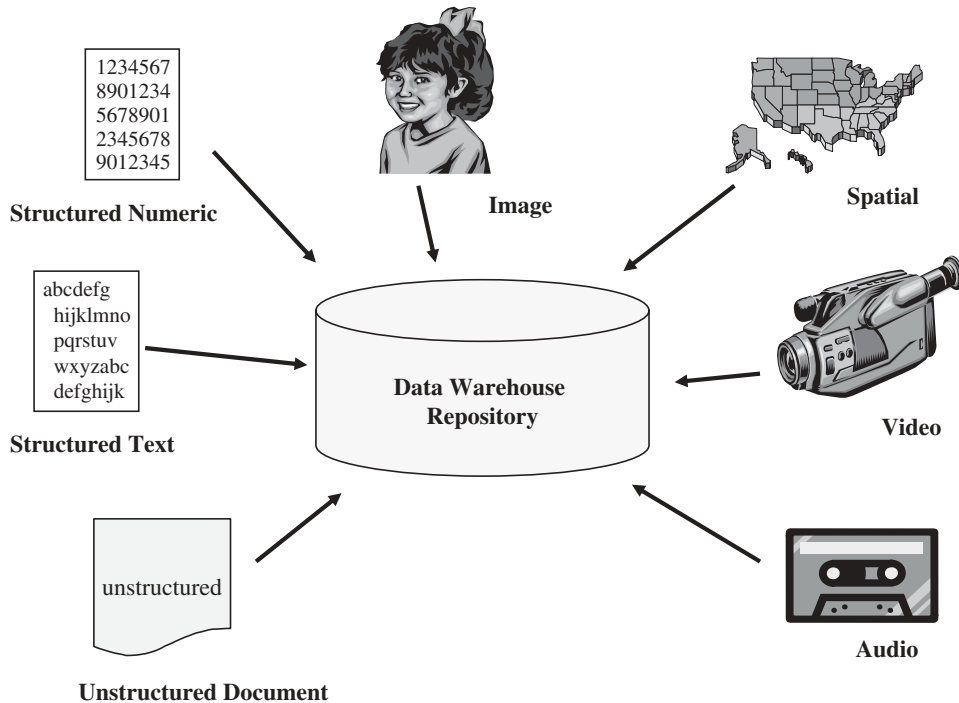


Figure 3-4 Multiple data types in a data warehouse.

Searching Unstructured Data You have enhanced your data warehouse by adding unstructured data. Is there anything else you need to do? Of course, without the ability to search unstructured data, integration of such data is of little value. Vendors are now providing new search engines to find the information the user needs from unstructured data. Query by image content is an example of a search mechanism for images. The product allows you to preindex images based on shapes, colors, and textures. When more than one image fits the search argument, the selected images are displayed one after the other.

For free-form text data, retrieval engines preindex the textual documents to allow searches by words, character strings, phrases, wild cards, proximity operators, and Boolean operators. Some engines are powerful enough to substitute corresponding words and search. A search with a word *mouse* will also retrieve documents containing the word *mice*.

Searching audio and video data directly is still in the research stage. Usually, these are described with free-form text, and then searched using textual search methods that are currently available.

Spatial Data Consider one of your important users, maybe the marketing director, being online and performing an analysis using your data warehouse. The marketing director runs a query: show me the sales for the first two quarters for all products compared to last year in store XYZ. After reviewing the results, he or she thinks of two other questions. What is the average income of people living in the neighborhood of that store? What is the average driving distance for those people to come to the store? These questions may be answered only if you include spatial data in your data warehouse.

Adding spatial data will greatly enhance the value of your data warehouse. Address, street block, city quadrant, county, state, and zone are examples of spatial data. Vendors have begun to address the need to include spatial data. Some database vendors are providing spatial extenders to their products using SQL extensions to bring spatial and business data together.

Data Visualization

When a user queries your data warehouse and expects to see results only in the form of output lists or spreadsheets, your data warehouse is already outdated. You need to display results in the form of graphics and charts as well. Every user now expects to see the results of a query shown as charts. Visualization of data in the result sets boosts the process of analysis for the user, especially when the user is looking for trends over time. Data visualization helps the user to interpret query results quickly and easily.

Major Visualization Trends In the last few years, three major trends have shaped the direction of data visualization software.

More Chart Types. Most data visualizations are in the form of some standard chart type. The numerical results are converted into a pie chart, a scatter plot, or another chart type. Now the list of chart types supported by data visualization software has grown much longer.

Interactive Visualization. Visualizations are no longer static. Dynamic chart types are themselves user interfaces. Your users can review a result chart, manipulate it, and then see newer views online.

Visualization of Complex and Large Result Sets. Your users can view a simple series of numeric result points as a rudimentary pie or bar chart. But newer visualization software can visualize thousands of result points and complex data structures.

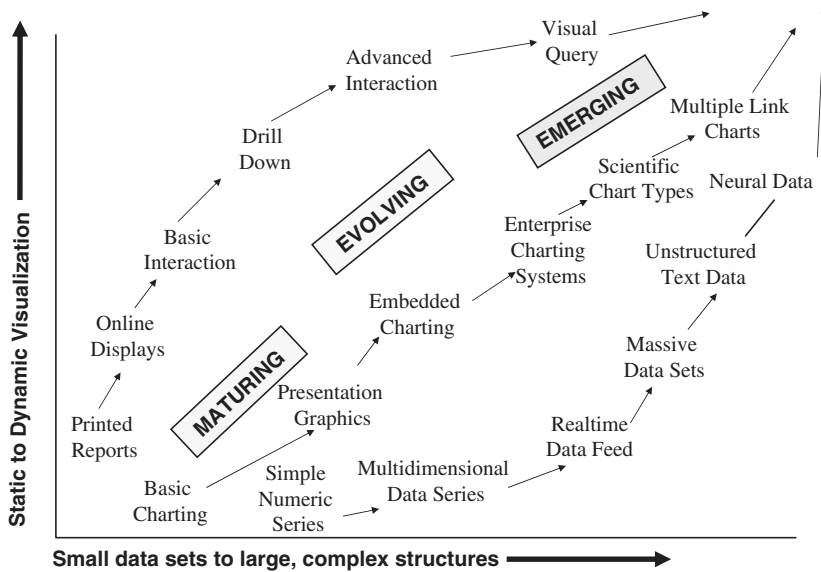


Figure 3-5 Data visualization trends.

Figure 3-5 summarizes these major trends. See how the technologies are maturing, evolving, and emerging.

Visualization Types Visualization software now supports a large array of chart types. Gone are the days of simple line graphs. The current needs of users vary enormously. Business users demand pie and bar charts. Technical and scientific users need scatter plots and constellation graphs. Analysts looking at spatial data need maps and other three-dimensional representations. In the last few years, major trends have shaped the direction of data visualization software.

Advanced Visualization Techniques The most remarkable advance in visualization techniques is the transition from static charts to dynamic interactive presentations.

Chart Manipulation. A user can rotate a chart or dynamically change the chart type to get a clearer view of the results. With complex visualization types such as constellation and scatter plots, a user can select data points with a mouse and then move the points around to clarify the view.

Drill Down. The visualization first presents the results at the summary level. The user can then drill down the visualization to display further visualizations at subsequent levels of detail.

Advanced Interaction. These techniques provide a minimally invasive user interface. The user simply double clicks a part of the visualization and then drags and drops representations of data entities. Or the user simply right clicks and chooses options from a menu. Visual query is the most advanced of user interaction features. For example, the user may see the outlying data points in a scatter plot, then select a few of them with the mouse and ask for a brand new visualization of just those selected points. The data visualization software generates the appropriate query from the selection, submits the query to the database, and then displays the results in another representation.

Dashboards and Scorecards Within the past few years many companies have adopted dashboards and scorecards as their preferred method of viewing performance information. In a certain sense, these means of obtaining information represent the culmination of business intelligence. Executives and managers, who need to monitor performance metrics, like digital dashboards that allow them to visualize the metrics as in speedometers, thermometers, or traffic lights. A dashboard informs users about what is happening; a scorecard informs users how well it is happening.

Dashboards monitor and measure processes. A dashboard provides real-time information like the dashboard in an automobile, enabling drivers to check current speed, engine temperature, fluid level, and so on. Dashboards are linked directly to real-time systems that capture events as they happen. Dashboards warn users with alerts or exception conditions. Industry vendors have improved dashboards to a very large extent and made them interactive and comprehensive.

Scorecards track progress compared to objectives. A scorecard displays periodic snapshots of performance viewed against an organization's strategic objectives and targets. Users select key performance indicators and have the progress displayed on the scorecards.

Parallel Processing

You know that the data warehouse is a user-centric and query-intensive environment. Your users will constantly be executing complex queries to perform all types of analyses. Each query would need to read large volumes of data to produce result sets. Analysis, usually performed interactively, requires the execution of several queries, one after the other, by each user. If the data warehouse is not tuned properly for handling large, complex, simultaneous queries efficiently, the value of the data warehouse will be lost. Performance is of primary importance.

The other functions for which performance is crucial are the functions of loading data and creating indexes. Because of large volumes, loading of data can be slow. Indexing in a data warehouse is usually elaborate because of the need to access the data in many different ways. Because of large numbers of indexes, index creation can also be slow.

How do you speed up query processing, data loading, and index creation? A very effective way to do this is to use parallel processing. Both hardware configurations and software techniques go hand in hand to accomplish parallel processing. A task is divided into smaller units and these smaller units are executed concurrently.

Parallel Processing Hardware Options In a parallel processing environment, you will find these characteristics: multiple CPUs, memory modules, one or more server nodes, and high-speed communication links between interconnected nodes.

Essentially, you can choose from three architectural options. Figure 3-6 indicates the three options and their comparative merits. Note the advantages and disadvantages so that you may choose the proper option for your data warehouse.

Parallel Processing Software Implementation You may choose the appropriate parallel processing hardware configuration for your data warehouse. Hardware alone would be worthless if the operating system and the database software cannot make use of the parallel features of the hardware. You will have to ensure that the software can allocate units of a larger task to the hardware components appropriately.

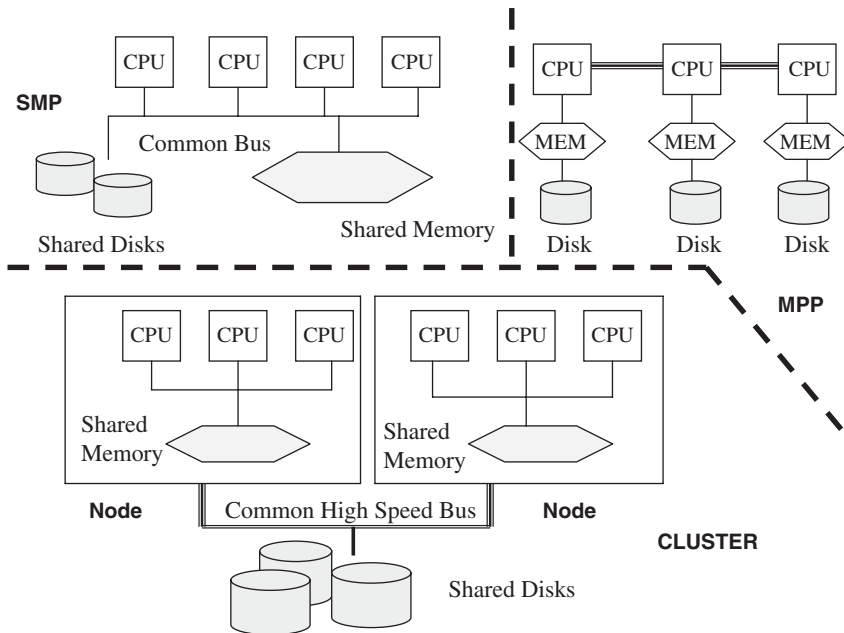


Figure 3-6 Hardware options for parallel processing.

Parallel processing software must be capable of performing the following steps:

- Analyzing a large task to identify independent units that can be executed in parallel
- Identifying which of the smaller units must be executed one after the other
- Executing the independent units in parallel and the dependent units in the proper sequence
- Collecting, collating, and consolidating the results returned by the smaller units

Database vendors usually provide two options for parallel processing: parallel server option and parallel query option. You may purchase each option separately. Depending on the provisions made by the database vendors, these options may be used with one or more of the parallel hardware configurations.

The parallel server option allows each hardware node to have its own separate database instance, and enables all database instances to access a common set of underlying database files. The parallel query option supports key operations such as query processing, data loading, and index creation.

Implementing a data warehouse without parallel processing options is almost unthinkable in the current state of the technology. In summary, you will realize the following significant advantages when you adopt parallel processing in your data warehouse:

- Performance improvement for query processing, data loading, and index creation
- Scalability, allowing the addition of CPUs and memory modules without changes to the existing application

- Fault tolerance so that the database is available even when some of the parallel processors fail
- A single logical view of the database even though the data may reside on the disks of multiple nodes

Data Warehouse Appliances

In the previous section we discussed how adoption of parallel processing has become a necessity in a data warehouse environment. More recently companies are realizing that fully adopting business intelligence could make or break the business. Vast quantities need to be retrieved for the decision-making process. Timeliness and depth of analysis have become crucial. Terabyte-sized data warehouses have become common.

Until recently vendors and users have handled this rapid growth in database size and complexity in analysis with very expensive and constant upgrading of hardware and software. Now it is becoming clearer that existing infrastructures are unable to handle the continued growth. The objective is to find a new approach to data warehousing infrastructure that is flexible, expandable, and specifically applicable to an organization. The result is the advent of data warehouse appliances.

The data warehouse appliance is designed specifically to take care of the workload of business intelligence. It is built with hardware and software components specifically architected for this purpose. Through its architecture, a data warehouse appliance integrates hardware, storage, and DBMS into one unified device. It combines the best elements of SMP and MPP to enable queries to be processed in the most optimal manner. Appliances from most vendors are designed to interface seamlessly with standardized applications and tools available on the business intelligence market.

A data warehouse appliance is quite scalable. Because all parts of the appliance, hardware and software, come from the same vendor, it is designed to be homogeneous and reliable. For the administrator, a data warehouse appliance provides simplicity because of its integrated nature.

Data warehouse appliances are already supporting data warehouse and business intelligence deployments at major corporations. This is especially true in telecommunications and retail where data volumes are enormous and queries and analysis more complex and demanding.

Query Tools

In a data warehouse, if there is one set of functional tools that is most significant, it is the set of query tools. The success of your data warehouse depends on your query tools. Because of this, data warehouse vendors have improved query tools during the past few years.

We will discuss query tools in greater detail in Chapter 14. At this stage, just note the following functions for which vendors have greatly enhanced their query tools.

- *Flexible presentation*—Easy to use and able to present results online and in reports, in many different formats.
- *Aggregate awareness*—Able to recognize the existence of summary or aggregate tables and automatically route queries to the summary tables when summarized results are desired.
- *Crossing subject areas*—Able to cross over from one subject data mart to another automatically.

- *Multiple heterogeneous sources*—Capable of accessing heterogeneous data sources on different platforms.
- *Integration*—Able to integrate query tools for online queries, batch reports, and data extraction for analysis, and provide seamless interface to go from one type of output to another.
- *Overcoming SQL limitations*—Able to provide SQL extensions to handle requests that cannot usually be done through standard SQL.

Browser Tools

Here we are using the term “browser” in a generic sense, not limiting it to Web browsers. Your users will be running queries against your data warehouse. They will be generating reports from your data warehouse. They will be performing these functions directly and not with the assistance of someone like you in IT. This is expected to be one of the major advantages of the data warehouse approach.

If the users have to go to the data warehouse directly, they need to know what information is available there. Users need good browser tools to browse through the informational meta-data and search to locate the specific pieces of information they want to receive. Similarly, when you are part of the IT team to develop your company’s data warehouse, you need to identify the data sources, the data structures, and the business rules. You also need good browser tools to browse through the information about the data sources. Here are some recent trends in enhancements to browser tools:

- Tools are extensible to allow definition of any type of data or informational object.
- Open APIs (application program interfaces) are included.
- Several types of browsing functions, including navigation through hierarchical groupings, are provided.
- Users are able to browse the catalog (data dictionary or metadata), find an informational object of interest, and proceed further to launch the appropriate query tool with the relevant parameters.
- Web browsing and search techniques are available to browse through the information catalogs.

Data Fusion

A data warehouse is a place where data from numerous sources is integrated to provide a unified view of the enterprise. Data may come from the various operational systems running on multiple platforms, where it may be stored in flat files or in databases supported by different DBMSs. In addition to internal sources, data from external sources is also included in the data warehouse. In the data warehouse repository, you may also find various types of unstructured data in the form of documents, images, audio, and video.

In essence, various types of data from multiple disparate sources need to be integrated or fused together and stored in the data warehouse. Data fusion is a technology dealing with the merging of data from disparate sources. It has a wider scope and includes real-time merging of data from instruments and monitoring systems. Serious research is being conducted in the technology of data fusion. The principles and techniques of data fusion technology have a direct application in data warehousing.

Data fusion not only deals with the merging of data from various sources, it also has another application in data warehousing. In present-day warehouses, we tend to collect data in astronomical proportions. The more information stored, the more difficult it is to find the right information at the right time. Data fusion technology is expected to address this problem also.

By and large, data fusion is still in the realm of research. Vendors are not rushing to produce data fusion tools yet. At this stage, all you need to do is to keep your eyes open and watch for developments.

Data Integration

In the previous section we touched upon the need for and challenges of combining data from disparate sources in a data warehouse. Integrating disparate data has been a challenge for data warehouse developers. This task is not becoming any easier given the data explosion in most businesses. Data integration has become a top priority and challenge for many companies. In this section we will consider the trends for integration being adopted in progressive organizations.

Data integration is not just confined to data warehousing projects, but is also important for companies to develop an enterprise-wide strategy for integration. The Data Warehousing Institute conducted a study on data integration approaches across the board. They noticed that large organizations are leaning towards an enterprise-wide integration architecture while medium-sized companies are dealing with data integration primarily from a business intelligence viewpoint.

Integration is viewed in a few different ways: Enterprise Information Integration (EII), Enterprise Application Integration (EAI), and so on. In general, integration can be brought about at four different levels in an information technology system: data integration, application integration, business process integration, and user interaction integration. An integration architecture must incorporate all these different levels.

Data Integration. The objective is to provide a single, unified view of the business data scattered across the various systems in an organization. This unified view may be a physical one that has been combined and consolidated from multiple disparate sources into a single repository such as a data warehouse or an operational data store. On the other hand, this may be a virtual federated view of disparate data pulled together dynamically at the time of data access. Another technique for data integration is the straightforward approach of merging data from one database to another, for example, merging customer data from a CRM database into a manufacturing system database.

Application Integration. Here the objective is to provide a unified view of business applications used in the organization. The flow of transactions, messages, and data between applications is controlled and managed to obtain a unified view.

Business Process Integration. The objective is to provide a unified view of the organization's business processes. This type of integration is achieved through business process design tools and business process management tools.

User Interaction Integration. The objective is to provide users with a single, personalized interface to the business processes and data for performing their organizational functions. This type of integration fosters data sharing and user collaboration. An enterprise portal is an example of a technique for user interaction integration.

Analytics

Data warehousing and business intelligence is all about analysis of data. During the past few years, vendors have made tremendous strides in providing effective tools for analysis. Companies are concentrating on two areas of analysis more than others.

Multidimensional Analysis Today, every data warehouse environment provides for multidimensional analysis. This is becoming an integral part of the information delivery system of the data warehouse. Provision of multidimensional analysis to your users simply means that they will be able to analyze business measurements in many different ways. Multidimensional analysis is also synonymous with online analytical processing (OLAP).

Because of the enormous importance of OLAP, we will discuss this topic in greater detail in Chapter 15. At this stage, just note that vendors have made tremendous progress in OLAP tools. Now vendor products are evaluated to a large extent by the strength of their OLAP components.

Predictive Analytics Already many organizations are reaping tremendous benefits by using predictive analytics in their business intelligence environment. Predictive analytics assist the organizations greatly by improving their understanding of their customer behaviors, by optimizing their business processes, by enabling them to anticipate problems before they arise, and by helping them to recognize opportunities well ahead of time.

Agent Technology

A software agent is a program that is capable of performing a predefined programmable task on behalf of the user. For example, on the Internet, software agents can be used to sort and filter out e-mail according to rules defined by the user. Within the data warehouse, software agents are beginning to be used to alert users to predefined business conditions. They are also beginning to be used extensively in conjunction with data mining and predictive modeling techniques. Some vendors specialize in alert system tools. You should definitely consider software agent programs for your data warehouse.

As the size of data warehouses continues to grow, agent technology gets applied more and more. Let us say your marketing analyst needs to use your data warehouse at specific, invariable intervals to identify threat and opportunity conditions that can offer business advantages to the enterprise. The analyst has to run several queries and perform multilevel analysis to find these conditions. Such conditions are exception conditions. So the analyst has to step through very intense iterative analysis. Some threat and opportunity conditions may be discovered only after long periods of iterative analysis. This takes up a lot of the analyst's time, perhaps on a daily basis.

Whenever a threat or opportunity condition is discovered through elaborate analysis, it makes sense to describe the event to a software agent program. This program will then automatically signal to the analyst every time that condition is encountered in the future. This is the very essence of agent technology.

Software agents may even be used for routine monitoring of business performance. Your CEO may want to be notified every time the corporate-wide sales fall below the monthly targets, three months in a row. A software agent program may be used to alert him or her every time this condition happens. Your marketing VP may want to know every time the monthly sales promotions in all the stores are successful. Again, a software agent program may be used for this purpose.

Syndicated Data

The value of the data content is derived not only from the internal operational systems, but from suitable external data as well. With the escalating growth of data warehouse implementations, the market for syndicated data is expanding rapidly.

Examples of traditional suppliers of syndicated data are A. C. Nielsen and Information Resources, Inc. for retail data and Dun & Bradstreet and Reuters for financial and economic data. Some of the earlier data warehouses incorporated syndicated data from such traditional suppliers to enrich the data content.

Now data warehouse developers are looking at a host of new suppliers dealing with many other types of syndicated data. The more recent data warehouses receive demographic, psychographic, market research, and other kinds of useful data from new suppliers. Syndicated data is becoming big business.

Data Warehousing and ERP

Look around to see what types of applications companies have been implementing in the last few years. You will observe a predominant phenomenon. Many businesses are adopting ERP (enterprise resource planning) application packages offered by major vendors like SAP, Baan, JD Edwards, and PeopleSoft. The ERP market is huge, crossing the \$50 billion mark.

Why are companies rushing into ERP applications? Most companies are plagued by numerous disparate applications that cannot present a single unified view of the corporate information. Many of the legacy systems are totally outdated. Reconciliation of data retrieved from various systems to produce meaningful and correct information is extremely difficult and, at some large corporations, almost impossible. Some companies were looking for alternative ways to circumvent the enormous undertaking of making old legacy systems Y2K-compliant. ERP vendors seemingly came to the rescue of such companies.

Data in ERP Packages A remarkable feature of an ERP package is that it supports practically every phase of the day-to-day business of an enterprise, from inventory control to customer billing, from human resources to production management, from product costing to budgetary control. Because of this feature, ERP packages are huge and complex. The ERP applications collect and integrate lots of corporate data. As these are proprietary applications, the large volumes of data are stored in proprietary formats available for access only through programs written in proprietary languages. Usually, thousands of relational database tables are needed to support all the various functions.

Integrating ERP and Data Warehouse In the 1990s and beyond, when ERP was introduced, this grand solution promised to bring about the integrated corporate data repositories companies were looking for. Because all data was cleansed, transformed, and integrated in one place, the appealing vision was that decision making and action taking could take place from one integrated environment. Soon companies implementing ERP realized that the thousands of relational database tables, designed and normalized for running the business operations, were not at all suitable for providing strategic information. Moreover, ERP data repositories lacked data from external sources and from other operational systems in the company. If your company has ERP or is planning to acquire ERP, you need to consider the integration of ERP with data warehousing.

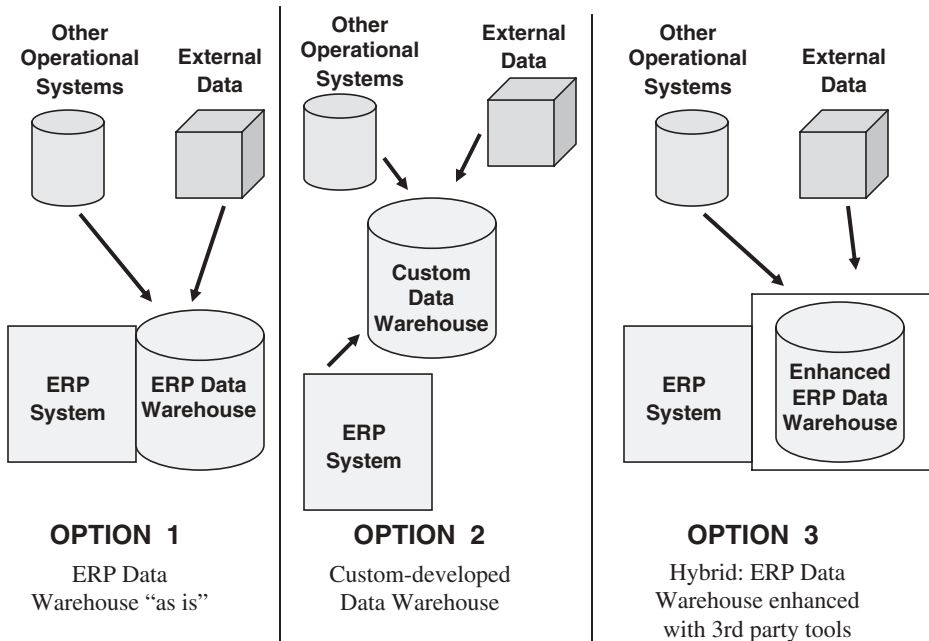


Figure 3-7 ERP and data warehouse integration options.

Integration Options Corporations integrating ERP and the data warehouse initiatives usually adopt one of three options shown in Figure 3-7. ERP vendors have begun to complement their packages with data warehousing solutions. Companies adopting option 1 implement the data warehousing solution of the ERP vendor with currently available functionality and await enhancements. The downside to this approach is that you may be waiting forever for the enhancements. In option 2, companies implement customized data warehouses and use third-party tools to extract data from the ERP datasets. Retrieving and loading data from proprietary ERP datasets is not easy. Option 3 is a hybrid approach that combines the functionalities provided by the vendor's data warehouse with additional functionalities from third-party tools.

You need to examine these three approaches carefully and pick the one most suitable for your corporation.

Data Warehousing and KM

If 1998 marked the resurgence of ERP systems, 1999 marked the genesis of knowledge management (KM) systems in many corporations. Since then, into the 2000s, KM caught on very rapidly. Operational systems deal with data; informational systems such as data warehouses empower users by capturing, integrating, storing, and transforming the data into useful information for analysis and decision making. Knowledge management takes this to a higher level. It completes the process by providing users with knowledge to use the right information, at the right time, and at the right place,

Knowledge is actionable information. What do we mean by *knowledge management*? It is a systematic process for capturing, integrating, organizing, and communicating knowledge accumulated by employees. It is a vehicle to share corporate knowledge so that employees

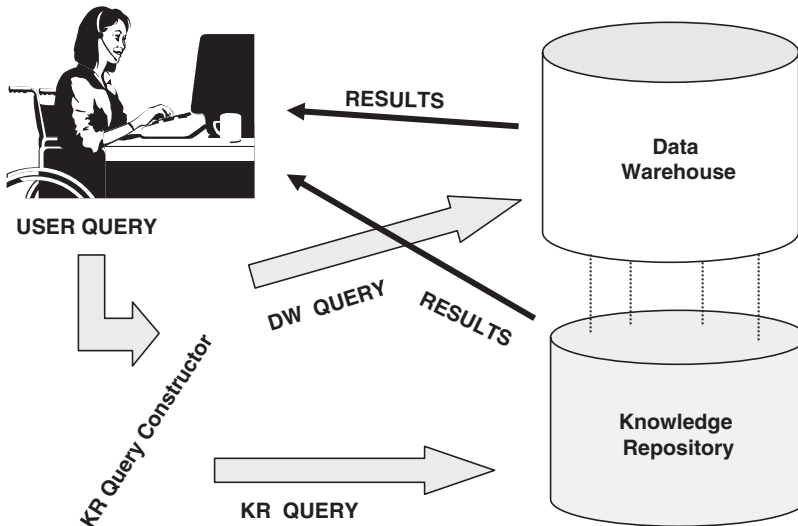
may be more effective and productive in their work. Where does the knowledge exist in a corporation? Corporate procedures, documents, reports analyzing exception conditions, objects, math models, what-if cases, text streams, video clips—all of these and many more such instruments contain corporate knowledge.

A knowledge management system must store all such knowledge in a knowledge repository, sometimes called a knowledge warehouse. A data warehouse contains structured information; a knowledge warehouse holds unstructured information. Therefore, a knowledge management framework must have tools for searching and retrieving unstructured information.

As a data warehouse developer, what are your concerns about knowledge management? Take a specific corporate scenario. Let us say sales have dropped in the south central region. Your marketing VP is able to discern this from your data warehouse by running some queries and doing some preliminary analysis. The vice president does not know why the sales are down, but things will begin to clear up if, just at that time, he or she has access to a document prepared by an analyst explaining why the sales are low and suggesting remedial action. That document contains the pertinent knowledge, although this is a simplistic example. The VP needs numeric information, but something more as well.

Knowledge, stored in a free unstructured format, must be linked to the sales results to provide context to the sales numbers from the data warehouse. With technological advances in organizing, searching, and retrieval of unstructured data, more knowledge philosophy will enter into data warehousing. Figure 3-8 shows how you can extend your data warehouse to include retrievals from the knowledge repository that is part of the knowledge management framework of your company.

Now, in the above scenario, the VP can get the information about the drop in sales from the data warehouse and then retrieve the relevant analyst's document from the knowledge repository. Knowledge obtained from the knowledge management system can provide



Integrated Data Warehouse—Knowledge Repository

Figure 3-8 Integration of KM and the data warehouse.

context to the information received from the data warehouse to understand the story behind the numbers.

Data Warehousing and CRM

Fiercer competition has forced many companies to pay greater attention to retaining customers and winning new ones. Customer loyalty programs have become the norm. Companies are moving away from mass marketing to one-on-one marketing. Customer focus has become the watchword. Concentration on customer experience and customer intimacy has become the key to better customer service. More and more companies are embracing customer relationship management (CRM) systems. A number of leading vendors offer turnkey CRM solutions that promise to enable one-on-one service to customers.

When your company is gearing up to be more attuned to high levels of customer service, what can you, as a data warehouse architect, do? If you already have a data warehouse, how must you readjust it? If you are building a new data warehouse, what are the factors for special emphasis? You will have to make your data warehouse more focused on the customer. You will have to make your data warehouse CRM-ready, not an easy task by any means. In spite of the difficulties, the payoff from a CRM-ready data warehouse is substantial.

Your data warehouse must hold details of every transaction at every touchpoint with each customer. This means every unit of every sale of every product to every customer must be gathered in the data warehouse repository. You need not only sales data in detail but also details of every other type of encounter with each customer. In addition to summary data, you have to load every encounter with every customer into the data warehouse. Atomic or detailed data provides maximum flexibility for the CRM-ready data warehouse. Making your data warehouse CRM-ready will increase the data volumes tremendously. Fortunately, today's technology facilitates large volumes of atomic data to be placed across multiple storage management devices that can be accessed through common data warehouse tools.

To make your data warehouse CRM-ready, you have to enhance some other functions also. For customer-related data, cleansing and transformation functions are more involved and complex. Before taking the customer name and address records to the data warehouse, you have to parse unstructured data to eliminate duplicates, combine them to form distinct households, and enrich them with external demographic and psychographic data. These are major efforts. Traditional data warehousing tools are not quite suited for the specialized requirements of customer-focused applications.

Agile Development

During the 1990s more and more organizations began to realize the problems encountered while adopting traditional methods of software development. The development methodology had been too structured, linear, and inflexible. Project delays and budget overruns became common and troublesome. Organizations suffered far-reaching consequences. Development efforts took too long and became too expensive. Users and stakeholders got frustrated. Ongoing feuds among groups in the development team resulted in rampant political maneuvers. Important issues slipped through the cracks. A radical new way of approaching software development was immediately called for.

Answering the clarion call for reformation, in February 2001 an initial group of 17 methodologists met in the city of Snowbird, Utah, and formed the Agile Software Development Alliance (briefly known as the Agile Alliance). The agile software development

methodology was born. This group did not consist of professionals with exactly the same skills and background; the group spanned a panel of professionals with varied backgrounds and experience levels. They met with one primary goal: to dramatically improve the software development effort.

The initial members of the Agile Alliance agreed on a bold manifesto that enumerates preferential and underlying themes for successful software development: people and interactions, usable software, user collaboration, and acceptance of change. Agile software development is based on a set of well-defined principles and practices. It is not a set of software tools and mechanisms for software development.

Today, many organizations have embraced the principles and practices of agile development and have realized remarkable success in their software projects. More recently companies have begun to apply agile development to data warehousing and business intelligence projects. In later chapters, we will discuss this further.

Active Data Warehousing

So far we have discussed a number of significant trends that are very relevant to what you need to bear in mind while building your data warehouse. Why not end our discussion of the significant trends with a bang? Let us look at what is known as active data warehousing.

What do you think of opening your data warehouse to 30,000 users worldwide, consisting of employees, customers, and business partners, in addition to allowing about 15 million users public access to the information every day? What do you think about making it a 24-7 continuous service delivery environment with 99.9% availability? Your data warehouse quickly becomes mission-critical instead of just being strategic. You are into active data warehousing.

This is what one global company has accomplished with an active data warehouse. The company operates in more than 60 countries, manufactures in more than 40 countries, conducts research in nearly 30 countries, and sells over 50,000 products in 200 countries. The advantages of opening up the data warehouse to outside parties other than employees are enormous. Suppliers work with the company on improved demand planning and supply chain management; the company and its distributors cooperate on planning between different sales strategies; customers make expeditious purchasing decisions. The active data warehouse truly provides one-on-one service to customers and business partners.

EMERGENCE OF STANDARDS

Think back to our discussion in Chapter 1 of the data warehousing environment as a blend of many technologies. A combination of multiple types of technologies is needed for building a data warehouse. The range is wide: data modeling, data extraction, data transformation, database management systems, control modules, alert system agents, query tools, analysis tools, report writers, and so on.

Now in a hot industry such as data warehousing, there is no scarcity of vendors and products. In each of the multitude of technologies supporting the data warehouse, numerous vendors and products exist. The implication is that when you build your data warehouse, many choices are available to you to create an effective solution with the best-of-breed products. That is the good news. However, the bad news is that when you try to use multivendor products, the result could also be total confusion and chaos. These multivendor products have to cooperate and work together in your data warehouse.

Unfortunately, there are no elaborate and established standards for the various products to exchange information and function together. When you use the database product from one vendor, the query and reporter tool from another vendor, and the OLAP (online analytical processing) product from yet another vendor, these three products have no standard method for exchanging data. Standards are especially critical in two areas: metadata interchange and OLAP functions.

Metadata is like the total roadmap to the information contained in a data warehouse. Each product adds to the total metadata content; each product needs to use metadata created by the other products. Metadata is like the glue that holds all the functional pieces together.

No modern data warehouse is complete without OLAP functionality. Without OLAP, you cannot provide your users full capability to perform multidimensional analysis, to view the information from many perspectives, and to execute complex calculations. OLAP is crucial.

In the following sections, we will review the progress made so far in establishing standards in these two significant areas. Although much progress has been made, as of the late 2000s, we have not achieved fully adopted standards in either of the areas.

Metadata

Two separate bodies have been working on standards for metadata, the Meta Data Coalition and the Object Management Group.

The Meta Data Coalition (MDC) was formed as a consortium of about 50 vendors and interested parties in October 1995 to launch a metadata standards initiative. The coalition worked on a standard known as the Open Information Model (OIM). Microsoft joined the coalition in December 1998 and became a staunch supporter along with some other leading vendors. In July 1999, the Meta Data Coalition accepted the Open Information Model as the standard and began to work on extensions. In November 1999, the coalition was driving new key initiatives. However, in November 2000, Meta Data Coalition discontinued its independent operations and merged with the Object Management Group. This paved the way for a unified standard combining the results of the efforts of two independent bodies.

The Object Management Group (OMG) comprised another group of vendors, including Oracle, IBM, Hewlett-Packard, Sun, and Unisys, which sought metadata standards through the Object Management Group, a larger, established forum dealing with a wider array of standards in object technology. In June 2000, the Object Management Group unveiled the Common Warehouse Metamodel (CWM) as the standard for metadata interchange for data warehousing.

In April 2000, the Meta Data Coalition and the Object Management Group announced that they would cooperate in reaching consensus on a single standard. This goal was realized in November 2000 when both bodies merged. As most corporate data is managed with tools from Oracle, IBM, and Microsoft, cooperation between the two camps was all the more critical. Since the merger, OMG has worked on integrating the Meta Data Coalition's OIM into its CWM.

OLAP

The OLAP Council was established in January 1995 as a customer advocacy group to serve as an industry guide. Membership and participation are open to interested organizations. In early 2000 the council included 16 general members, mainly vendors of OLAP products.

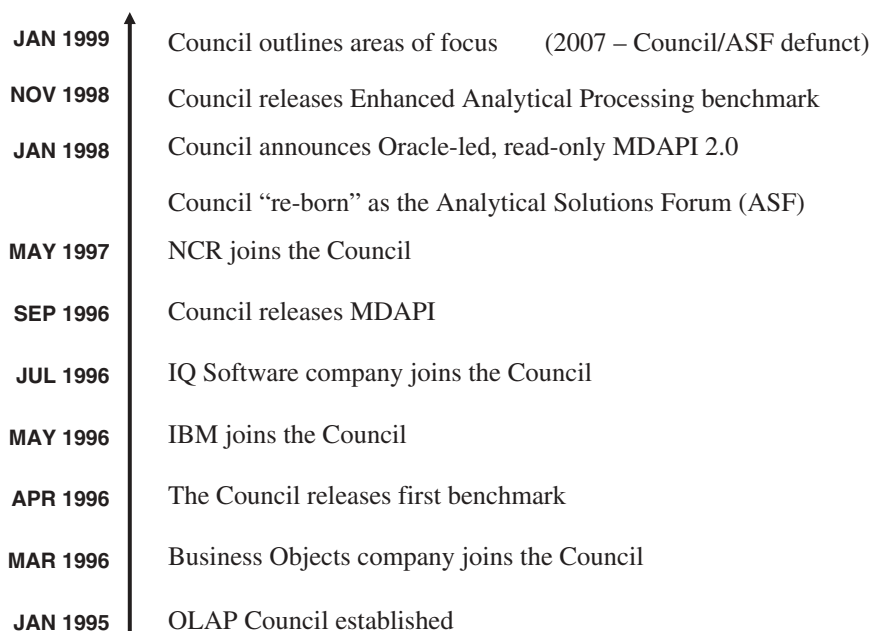


Figure 3-9 OLAP Council: activities timeline.

Over the years, the council has worked on OLAP standards for the Multi-Dimensional Application Programmers Interface (MDAPI) and has come up with revisions. Figure 3-9 shows a timeline of the earlier activities of the council.

Several OLAP vendors, platform vendors, consultants, and system integrators had indicated their support for MDAPI 2.0. OLAP membership dwindled and soon the council was reborn as the Analytical Solutions Forum (ASF). In practice, MDAPI 2.0 never really got adopted by any substantial group of vendors. It appears the OLAP Council and its successor ASF have been long forgotten.

WEB-ENABLED DATA WAREHOUSE

We all know that the single most remarkable phenomenon that has impacted computing and communication during the last few years is the Internet. At every major industry conference and in every trade journal, most of the discussions relate to the Internet and the World Wide Web in one way or another.

Starting with a meager number of just four host computer systems in 1969, the Internet swelled to gigantic proportions, with nearly 95 million hosts by 2000. It is still growing exponentially. The number of World Wide Web sites escalated to nearly 26 million by 2000. Nearly 150 million global users get on the Internet. Making full use of the ever-popular Web technology, numerous companies have built Intranets and Extranets to reach their employees, customers, and business partners. The Web has become the universal information delivery system.

We are also aware of how the Internet has fueled the tremendous growth of electronic commerce in recent years. Annual volume of business-to-business e-commerce exceeds

\$300 billion and total e-commerce will soon pass the \$1 trillion mark. No business can compete or survive without a Web presence. The number of companies conducting business over the Internet had grown to 400,000 by 2003.

As a data warehouse professional, what are the implications for you? Clearly, you have to tap into the enormous potential of the Internet and Web technology for enhancing the value of your data warehouse. Also, you need to recognize the significance of e-commerce and enhance your warehouse to support and expand your company's e-business.

You have to transform your data warehouse into a Web-enabled data warehouse. On the one hand, you have to bring your data warehouse to the Web, and, on the other hand, you need to bring the Web to your data warehouse. In the next two subsections, we will discuss these two distinct aspects of a Web-enabled data warehouse.

The Warehouse to the Web

In early implementations, the corporate data warehouse was intended for managers, executives, business analysts, and a few other high-level employees as a tool for analysis and decision making. Information from the data warehouse was delivered to this group of users in a client-server environment. But today's data warehouses are no longer confined to a select group of internal users. Under present conditions, corporations need to increase the productivity of all the members in the corporation's value chain. Useful information from the corporate data warehouse must be provided not only to employees but also to customers, suppliers, and all other business partners.

So in today's business climate, you need to open your data warehouse to the entire community of users in the value chain, and perhaps also to the general public. This is a tall order. How can you accomplish this requirement to serve information to thousands of users in 24-7 mode? How can you do this without incurring exorbitant costs for information delivery? The Internet, along with Web technology, is the answer. The Web will be your primary information delivery mechanism.

This new delivery method will radically change the ways your users will retrieve, analyze, and share information from your data warehouse. The components of your information delivery will be different. The Internet interface will include a browser, search engine, push technology, home page, information content, hypertext links, and downloaded Java or ActiveX applets.

When you bring your data warehouse to the Web, from the point of view of the users, the key requirements are: self-service data access, interactive analysis, high availability and performance, zero-administration client (thin client technology such as Java applets), tight security, and unified metadata.

The Web to the Warehouse

Bringing the Web to the warehouse essentially involves capturing the clickstream of all the visitors to your company's Web site and performing all the traditional data warehousing functions. And you must accomplish this, near real-time, in an environment that has now come to be known as the data Webhouse. Your effort will involve extraction, transformation, and loading of the clickstream data to the Webhouse repository. You will have to build dimensional schemas from the clickstream data and deploy information delivery systems from the Webhouse.

Clickstream data tracks how people proceeded through your company's Web site, what triggers purchases, what attracts people, and what makes them come back. Clickstream data enables analysis of several key measures, including:

- Customer demand
- Effectiveness of marketing promotions
- Effectiveness of affiliate relationship among products
- Demographic data collection
- Customer buying patterns
- Feedback on Web site design

A clickstream Webhouse may be the single most important tool for identifying, prioritizing, and retaining e-commerce customers. The Webhouse can produce the following useful information:

- Site statistics
- Visitor conversions
- Ad metrics
- Referring partner links
- Site navigation resulting in orders
- Site navigation not resulting in orders
- Pages that are session killers

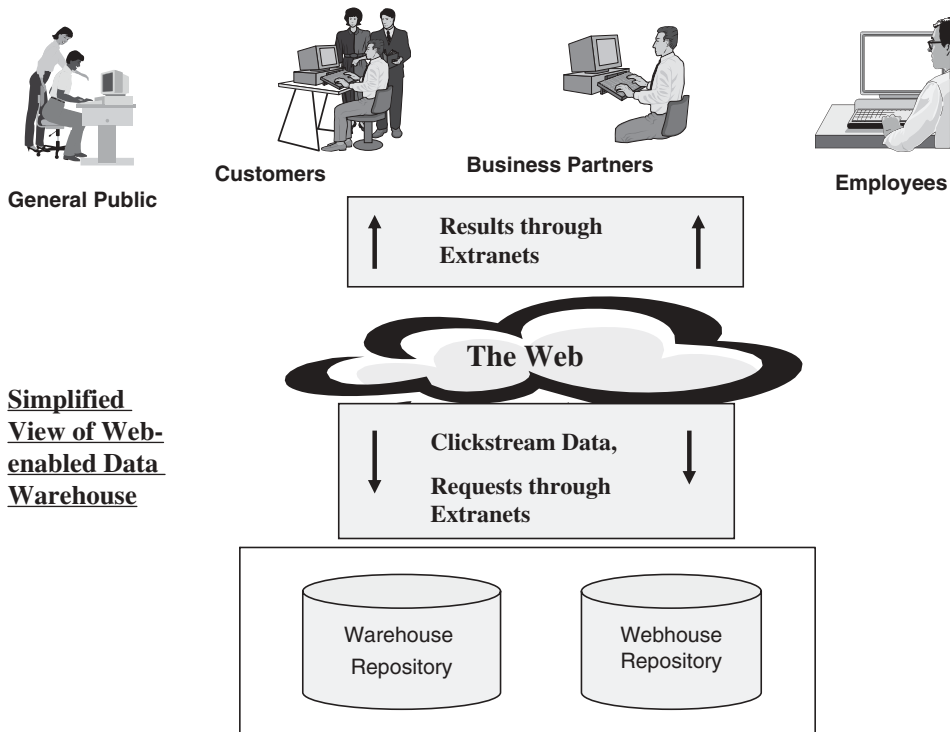


Figure 3-10 Web-enabled data warehouse.

- Relationships between customer profiles and page activities
- Best customer and worst customer analysis

The Web-Enabled Configuration

Figure 3-10 indicates an architectural configuration for a Web-enabled data warehouse. Notice the presence of the essential functional features of a traditional data warehouse. In addition to the data warehouse repository holding the usual types of information, the Webhouse repository contains clickstream data.

The convergence of the Web and data warehousing is of supreme importance to every corporation doing business in the twenty-first century. Because of its critical significance, we will discuss this topic in much greater detail in Chapter 16.

CHAPTER SUMMARY

- Data warehousing has become mainstream with the spread of high-volume data warehouses and the rapid increase in the number of vendor products.
- To be effective, modern data warehouses need to store multiple types of data: structured and unstructured, including documents, images, audio, and video.
- Data visualization deals with displaying information in several types of visual forms: text, numerical arrays, spreadsheets, charts, graphs, and so on. Tremendous progress has been made in data visualization, including information through dashboards and scorecards.
- Data warehouse performance may be improved by using parallel processing with appropriate hardware and software options. Many companies are opting for data warehouse appliances.
- It is critical to adapt data warehousing to work with ERP packages, knowledge management, and customer relationship systems.
- The data warehousing industry is seriously seeking agreed-upon standards for metadata and OLAP. The end is perhaps in sight.
- Web-enabling the data warehouse means using the Web for information delivery and integrating the clickstream data from the corporate Web site for analysis. The convergence of data warehousing and Web technology is crucial to every business in the twenty-first century.

REVIEW QUESTIONS

1. State any three factors that indicate the continued growth in data warehousing and business intelligence. Can you think of some examples?
2. Why do data warehouses continue to grow in size, storing huge amounts of data? Give any three reasons.
3. Why is it important to store multiple types of data in the data warehouse? Give examples of some nonstructured data likely to be found in the data warehouse of a health management organization (HMO).
4. What is meant by data fusion? Where does it fit in data warehousing?

5. Describe four types of charts you are likely to see in the delivery of information from a data mart supporting the finance department.
6. What is your understanding of data warehouse appliances? List some of the benefits of data warehouse appliances.
7. Explain what is meant by agent technology? How can this technology be used in a data warehouse?
8. Describe any one of the options available to integrate ERP with data warehousing.
9. What is CRM? How can you make your data warehouse CRM-ready?
10. What do we mean by a Web-enabled data warehouse? Describe three of its functional features.

EXERCISES

1. Indicate if true or false:
 - A. Data warehousing helps in customized marketing.
 - B. It is as important to include unstructured data as structured data in a data warehouse.
 - C. Dynamic charts are themselves user interfaces.
 - D. MPP is a shared-memory parallel hardware configuration.
 - E. ERP systems may be substituted for data warehouses.
 - F. Most of a corporation's knowledge base contains unstructured data.
 - G. The traditional data transformation tools are quite adequate for a CRM-ready data warehouse.
 - H. Metadata standards facilitate deploying a combination of best-of-breed products.
 - I. MDAPI is a data fusion standard.
 - J. A Web-enabled data warehouse stores only the clickstream data captured at the corporation's Web site.
2. As the senior analyst on the data warehouse project of a large retail chain, you are responsible for improving data visualization of the output results. Make a list of your recommendations.
3. Explain how and why parallel processing can improve performance for data loading and index creation.
4. Discuss three specific ways in which agent technology may be used to enhance the value of the data warehouse in a large manufacturing company.
5. Your company is in the business of renting DVDs and video tapes. The company has recently entered into e-business and the senior management wants to make the existing data warehouse Web-enabled. List and describe any three of the major tasks required for satisfying the management's directive.